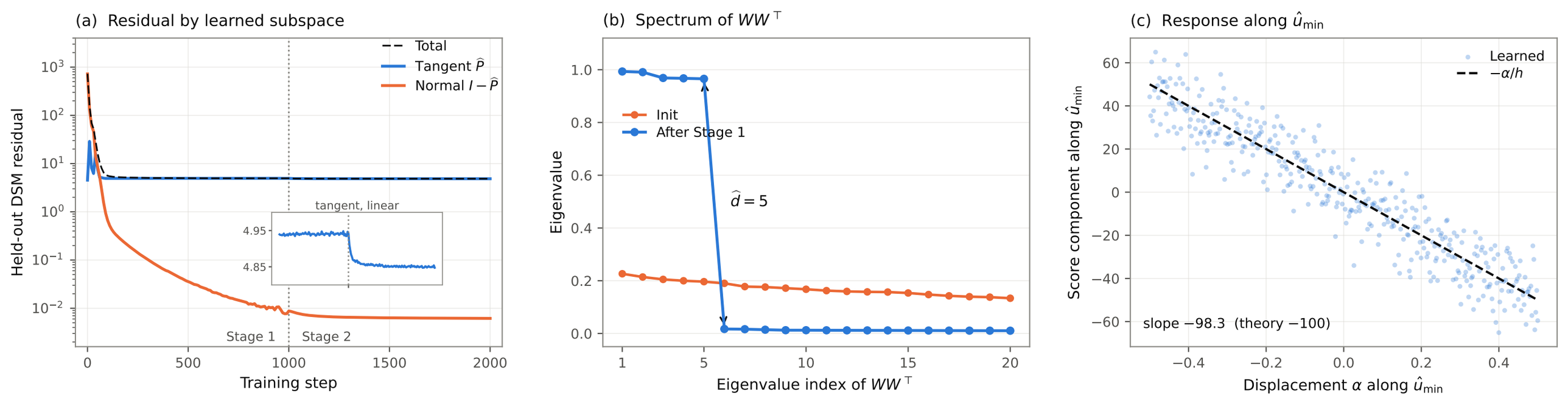
(a) Residual by learned subspace
Total
Tangent $\widehat{P}$
Normal $I-\widehat{P}$
Held-out DSM residual
10^{3}
10^{2}
10^{1}
10^{0}
10^{-1}
10^{-2}
tangent, linear
4.95
4.85
Stage 1
Stage 2
0
500
1000
1500
2000
Training step
(b) Spectrum of $WW^{\top}$
Init
After Stage 1
$\widehat{d}=5$
Eigenvalue
1.0
0.8
0.6
0.4
0.2
0.0
1
5
10
15
20
Eigenvalue index of $WW^{\top}$
(c) Response along $\hat{u}_{\min}$
Learned
$-\alpha/h$
Score component along $\hat{u}_{\min}$
60
40
20
0
−20
−40
−60
slope −98.3 (theory −100)
−0.4
−0.2
0.0
0.2
0.4
Displacement α along $\hat{u}_{\min}$